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SPD5118

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SPD HUB for DDR5 Memory Modules

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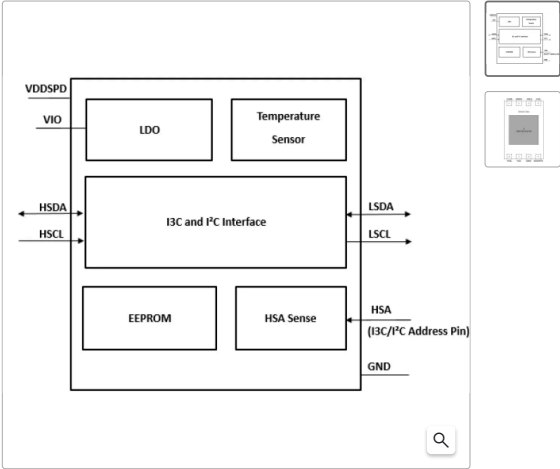
Support

Features

- Two-wire programmable I²C or I²C bus serial interface
- Single device load on the Host bus
- Up to 12.5MHz transfer rate
- 1.8V power supply input
- 1.0V VLDO output at VIO pin; optionally can take 1.0V input power supply on VIO pin
- Supports 1.0V, 1.1V, and 1.2V Push-Pull IO levels
- Supports 1.0V, 1.1V, 1.2V, 1.8V, 2.5V and 3.3V Open Drain IO levels
- 16 blocks of non-volatile memory (NVM), 64 bytes per block
- Hub function with 3 address bit translation
- Integrated temperature sensor; 0.5 °C accuracy with 0.25 °C resolution
- Packet Error Check (PEC) function
- Parity Error Check function
- Up to 8 unique addressing
- Programmable I²C, I²C bus addressing scheme
- In Band Interrupt (IBI)
- Write protection for each block of NVM
- 9-pin thermally enhanced DFN package
- Temperature: Industrial range of -40 °C to 125 °C

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SPD5118 - Block Diagram

Description

The SPD5118 hub device is a DDR5 Serial Presence Detect (SPD) EEPROM with Hub function (SPD5 Hub) and integrated temperature sensor (optional) as used for memory module applications. The hub feature allows for the isolation of a local bus from a master host bus. This SPD5 device contains 1024 bytes of non-volatile memory (EEPROM) arranged as 16 blocks of 64 bytes per block. Each block may optionally be write-protected via software command.

Parameters

Attributes	Value
Product Type	SPD Hub
Function	SPD Hub for DDR5 Memory Modules
Application System	Server and Client
Input Voltage Range (V)	3 - 3.6

Package Options

Pkg. Type	Pkg. Dimensions (mm)	Lead Count (#)	Pitch (mm)
DFN	2.0 x 3.0 x 0.75	8	0.5

Application Block Diagrams

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